Comparative Analysis of Influenza A and SARS-CoV-2 Transmission Dynamics Using Agent-Based Modelling

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1 Introduction

Every new outbreak of viral infections presents a major global public health threat. The primary objective of this study is to learn more about the dynamics of transmission of SARS-CoV-2 and influenza A, the two most prevalent respiratory viruses. These viruses, responsible for COVID-19 and seasonal flu respectively, have had profound effects on global health systems and societies. With the intention of developing strategies to slow their spread, it is essential to understand how they propagate, including the effectiveness of mitigation strategies such as wearing masks and quarantine.

The framework of this coursework assignment is agent-based modelling, a powerful approach in epidemiology for simulating interactions within populations to study the spread of diseases. This research aims to provide insights into how these viruses spread, especially through changes in individual health metrics, through the creation of virtual environments and agents that replicate human behaviour and the typical course of virus transmission. Furthermore, this project includes an exploration of how the implementation of mitigation strategies like mask-wearing and quarantine can alter transmission dynamics.

This project specifically aims to:

- Create two distinct agent-based models to replicate the transmission settings of SARS-CoV-2 and influenza A.
- Examine how these viruses spread in diverse situations that are impacted by the people' varying health characteristics.
- Assess the impact of variables including incubation and infectious periods, asymptomatic proportions, and the adoption of mitigation strategies such as mask-wearing and quarantine on virus transmission.

This examination will be guided by the following research questions:

1. In a simulated setting, how do the transmission dynamics of Influenza A and SARS-CoV-2 differ?
2. What are the peak times of infection for each virus under varying conditions of individual health parameters and mitigation strategies?

In addition to adding to our knowledge of epidemiological modelling, this work provides insights that may be applied to health management policies. By analysing these simulations, we aim to identify how individual health variations and mitigation strategies affect the transmission of these infectious diseases, thus aiding in the planning and response to future outbreaks.

2 Literature Review

This literature review explores how recent developments in agent-based models (ABMs) have improved our understanding and ability to simulate the dynamics of infectious diseases. The examined works cover basic ideas first proposed by pioneers in the field, comprehensive analyses of the last ten years of progress, and the most recent approaches used in response to modern pandemics such as Covid-19.

2.1 Foundational Concepts in Epidemiology

Anderson and May's book, reviewed by Tillett (1992), remains a cornerstone in the epidemiological study of infectious diseases. It presents an organised analysis of the dynamic regulation and transmission mechanisms of infectious illnesses, offering a theoretical framework for current modelling endeavours.

2.2 Systematic Reviews of Agent-Based Models

Willem et al. (2017) provide a systematic review of a decade of individual-based models, evaluating the development of this methodology in the field of disease transmission research. Their assessment demonstrates the great advancements in simulation accuracy and application to public health policy, as well as the adaptability of ABMs to a wide range of infectious diseases.

2.3 Agent-Based Models in Recent Pandemics

2.3.1 COVID-19 Modelling

Chang et al. (2020) demonstrated how different public health initiatives could affect disease transmission by simulating the spread of COVID-19 in Australia using ABMs. This study demonstrates the practical application of ABMs in real-time pandemic response and policy-making.

2.3.2 Methodological Reviews

In their methodological review of the epidemic models employed in the first wave of the COVID-19 pandemic, Garin et al. provide a critical evaluation of the efficacy of different modelling techniques in forecasting the development of the disease.

2.3.3 Covasim Tool

Kerr et al. (2021) developed Covasim, an agent-based model for COVID-19, to assess the effects of interventions such as vaccinations and social distancing. This tool has been crucial in providing insights into the dynamics of disease spread and intervention outcomes.

2.3.4 High-Resolution Modelling

Truszkowska et al. (2021) used high-resolution ABMs to study COVID-19 spread in a small town, showcasing the model's ability to represent intricate interactions and its significance for comprehending the dynamics of confined outbreaks.

2.4 Recent Studies on Vaccination and Public Health Strategies

- **Vaccination strategies:** Using an agent-based model, Singh et al. (2022) assessed several vaccination strategies in Madrid. Their research helps to improve vaccination campaigns and customize policies to the capabilities of the health system and specific population dynamics.
- **Survey of Models:** Alsharhan (2021) provides a survey of agent-based simulations specifically for modelling the COVID-19 pandemic, emphasizing how flexible ABMs are in the face of quickly changing epidemiological environments and how important a role they play in strategic planning.

2.5 Conclusion

The literature review highlights the remarkable progress made in agent-based modelling, ranging from basic ideas to complex simulations designed for particular outbreaks such as COVID-19. These studies demonstrate the advancements in modelling methods while also highlighting the vital role that ABMs play in developing successful public health interventions. The prediction capability and practicality of these models in the management of infectious diseases will be further improved as the area develops through the integration of real-time data, machine learning, and more precise individual-level factors.

3 Environment and Agent Design

The design of both the environment and agents is crucial for accurately simulating the transmission dynamics of Influenza A and SARS-CoV-2, including the evaluation of mitigation strategies like mask-wearing and quarantine. The environment in this project is a virtual public area where agents, who stand in for members of a population, engage in ways that are similar to those of real-world public interactions.

3.1 Environment

The simulated environment is a public space designed to reflect common areas such as parks, markets, and public transport systems. These environments are picked because of their high risk of virus transmission because they involve lots of people interacting with each other frequently and in a variety of ways. The risk in these environments makes them ideal for studying the effects of mitigation strategies like mask-wearing and movement restrictions on disease spread. In the context of this simulation, quarantine is specifically implemented by reducing the movement of infected agents within the public space.

3.2 Agents

The characteristics of agents are modelled to reflect physiological and behavioural characteristics of humans that are important in the transmission of disease. Important characteristics consist of:

- **Health Status:** The state of each agent's health—defined as "susceptible," "infected," or "recovered"—affects how they interact with one another and move around in the environment.
- **Infectiousness:** Infected agents have a window of time during which they can spread the virus to nearby susceptible agents. For the simulation to work, the length of this interval and the transmission probability per touch are essential.
- **Symptom Presentation:** Agents may display a range of symptoms from asymptomatic to severe, affecting their likelihood to be in public spaces and their interaction frequency.
- **Health Parameters:** An agent's susceptibility to infection and the course of the disease are influenced by factors like age, the strength of their immune system, and pre-existing medical disorders.

3.3 Simulation Dynamics

The simulation operates on discrete time steps, each representing one day. During these steps, agents move between health states according to illness progression models and interact with each other based on proximity. Mitigation strategies are directly integrated into the simulation as follows:

- **Mask-Wearing:** This strategy is modelled by reducing the infectious radius of agents who are wearing masks, thus decreasing the likelihood of transmitting the virus to others.
- **Quarantine:** Quarantine measures are simulated by restricting the movement of "infected" agents, which prevents them from coming into contact with other agents and spreading the infection further.

This focused approach makes it possible for the simulation to accurately represent the dynamics of disease transmission in public environments and offers a strong foundation for researching how different health characteristics and specific mitigation strategies affect the spread of respiratory viruses.

4 Implementation

The design of the agent-based models for SARS-CoV-2 and influenza A was structured to simulate the dynamics of the viruses' propagation in a public setting. This section includes a quick summary of the main elements of the implementation as well as the technical setup, tools, and libraries used.

4.1 Technical Setup and Tools

Python was used to create the simulations because of its strong ecosystem and abundance of packages that are well-suited for computational modelling. Important libraries that are being utilised are:

- **NumPy:** For efficient numerical operations, especially array manipulations which are fundamental in handling the state of each agent at every simulation timestep.
- **Pandas:** Employed for data management and analysis, facilitating the handling of time-series data for infection rates and other epidemiological metrics.
- **Matplotlib and Seaborn:** For visualisation of results, these libraries provided the tools to create dynamic graphs and charts that illustrate the transmission trends and outcomes of various scenarios.

4.2 Key Components

The following interrelated components make up the core of the implementation:

- **Agent Class:** Each agent is instantiated as an object that possesses attributes such as health status, infectiousness, symptom presentation, and individual health parameters. Methods within this class manage status transitions (e.g., from susceptible to infected) based on interactions and disease progression. Additionally, agents have a mask attribute indicating whether they are wearing a mask, which affects their infectious radius, reducing the likelihood of transmission.

- **Environment Setup:** An agent's interactions take place within a grid-like or network-like environment. In order to replicate the dynamics of a public area, this configuration involves establishing the spatial factors that affect how agents move and assemble. The environment setup includes mechanisms to enforce quarantine measures selectively, restricting movement for certain agents to simulate real-world social distancing and isolation protocols.

- **Transmission Model:** This model incorporates up-to-date scientific knowledge about the viruses, including transmission rates and the impacts of symptomatic and asymptomatic infections. The model simulates the probability of disease transmission based on agent proximity and interaction duration, with modifications to account for mitigative actions like mask-wearing and quarantine.

- **Simulation Engine:** Drives the progression of the simulation over defined time steps. It updates the state of each agent and the overall environment at each step, applying the rules for movement and interaction based on the agents' health status and the environment's characteristics. It is equipped to apply mitigation strategies such as mask-wearing and quarantine at specific points.

- **Mitigation Strategies:** Embedded within the simulation are checkpoints or triggers based on epidemiological thresholds where mitigation strategies are activated. These include the mandatory wearing of masks and the imposition of quarantine measures. Agents respond to these directives based on their attributes and the simulated policy environment, altering their behaviour to facilitate a detailed analysis of strategy effectiveness.

- **Data Collection and Analysis Module:** This component automatically records data points at each timestep, such as the number of infected agents, the spread pattern, and recovery rates.

4.3 Code Example

Below is a pseudocode snippet illustrating the agent interaction logic:

```
for each agent in simulation:
    if agent.health_status == 'infected' and agent.day_since_infection
    < infectious_period:
        for each nearby_agent in agent.get_nearby_agents():
            if nearby_agent.health_status == 'susceptible':
                if random() < transmission_probability:
                    nearby_agent.health_status = 'infected'
```

Listing 1: Disease transmission through proximity

This pseudocode reflects the fundamental mechanism through which the disease is transmitted in the simulation, emphasizing the probabilistic nature of infection based on proximity and health status.

4.4 Challenges

Realistic agent behaviour modelling and simulation optimisation for large-scale runs were two of the implementation's challenges. These were resolved by testing and improving the transmission models and interaction rules iteratively, which improved the simulations' accuracy and performance.

5 Experimental Setup

In a simulated public setting, the experimental framework for this study was created to evaluate the dynamics of SARS-CoV-2 and influenza A transmission under various agent health and population density scenarios. The method involves gathering data in a methodical manner using simulations and then analysing it to investigate how different health characteristics and partial compliance with mitigation measures affect the course of diseases.

5.1 Data Collection

Utilising the simulation environment, a Python script was used to automate the data collection experiments. The setup involves varying the type of virus, number of agents, and proportion of agents with comorbidities across multiple iterations to ensure robust statistical analysis. Specifically:

- **Virus Types:** Influenza and SARS-CoV-2, allowing comparison of their transmission characteristics.
- **Number of Agents:** Ranging from 200 to 2000, this parameter helps understand the impact of population density on virus spread.
- **Comorbidity Proportions:** From 0.1 to 0.5, to study how varying health vulnerabilities within a population affect the disease dynamics.
- **Mitigation Strategies:** Measures like mask wearing and quarantine are tested, with the *mitigation_population* parameter defining the percentage of the population that complies with these rules.
- **Iterations:** Each configuration is repeated 20 times to minimise random fluctuations in the simulation outcomes.

The results from these simulations are stored in a CSV file with detailed records for each run, including metrics like days to the end of the outbreak, peak infection day, and total numbers of infected, dead, and recovered agents.

5.2 Data Analysis

The collected data are analysed using Python with Pandas and Matplotlib for processing and visualisation:

- **Average Total Dead:** Compares the mortality across virus types and comorbidity proportions, providing insights into the lethality of each virus under different health conditions.
- **Peak Infection Trends:** Analyses how the peak infection count varies with comorbidity and virus type, indicating the potential for widespread transmission.
- **Duration of Outbreak:** Explores the duration of the outbreak in relation to the number of agents and their health statuses, which helps gauge the outbreak's containment under varied scenarios.
- **Mortality Rates:** Examines the death rates relative to the total population and comorbidity proportions, offering a perspective on risk assessment for public health strategies.
- **Recovery Trends:** Assesses recovery rates in correlation with population density and health variability, which could influence recovery phase planning in real-world scenarios.

5.3 Statistical Tools

Statistical analysis primarily utilises aggregation functions like mean and standard deviation within pivot tables, and visualisations include bar charts, line plots, and area charts to illustrate the findings comprehensively. These tools facilitate a deep understanding of the data, highlighting significant patterns and anomalies that address the research questions.

6 Results and Analysis

The simulation results provide light on how different mitigation methods can alter the dynamics of virus transmission. The simulations were created to evaluate the effects of different strategies on the spread of SARS-CoV-2 and Influenza A in a simulated public environment. The data has been visualised to demonstrate the effectiveness of masks, quarantine, and a combination of both in containing the spread of these viruses. The data was gathered from the simulation using Python scripts.

6.1 Mortality Rate Analysis

The first set of graphs presents the mortality rates based on comorbidity proportions and mitigation strategies (see Figure 1). The findings show that larger levels of comorbidity are associated with higher fatality rates, which are largely offset by mask and quarantine use together. For populations with greater health vulnerabilities, this impact emphasises the significance of focused protective interventions.

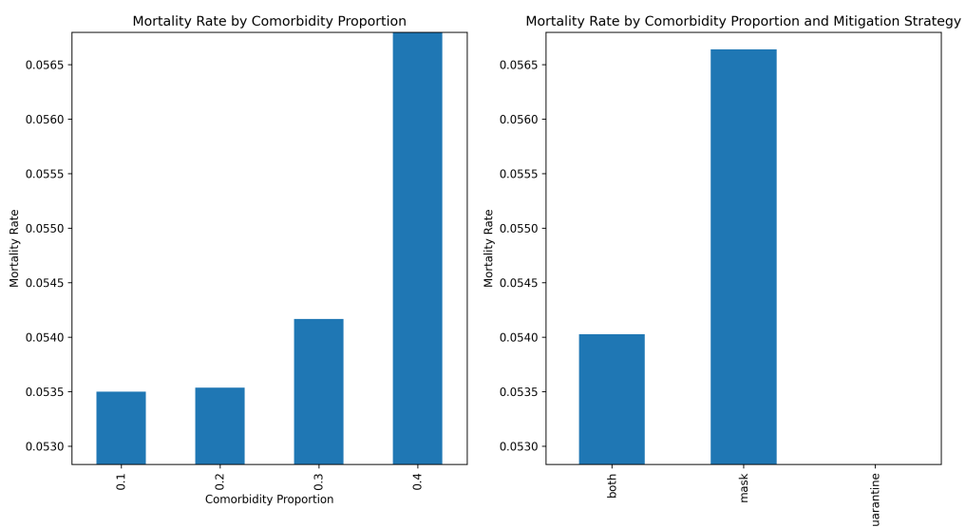

Figure 1: Mortality rates based on comorbidity proportions and mitigation strategies.

6.2 Total Infected by Mitigation Strategy

The second graph illustrates the average total number of infected individuals under different mitigation strategies (see Figure 2). The three strategies include masks alone, quarantine alone, and both combined. When applied consistently throughout the population, the data show little variation in the average total infections among the techniques, indicating a similar level of success in lowering the overall number of illnesses.

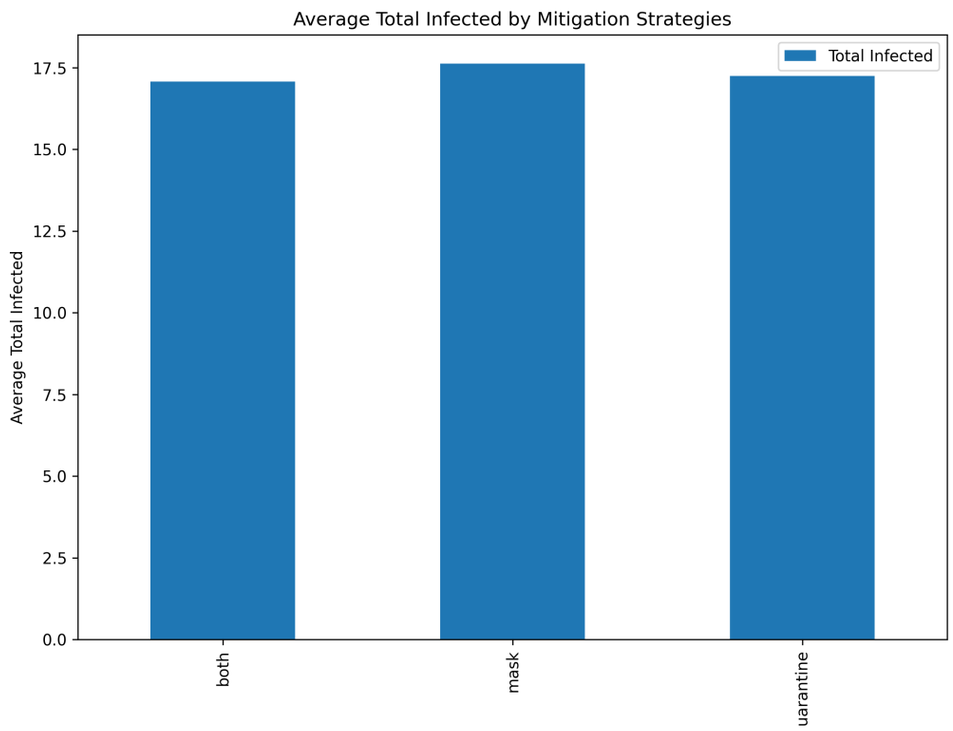

Figure 2: Average total number of infected individuals under different mitigation strategies.

6.3 Duration of Outbreak by Mitigation Strategy

The third graph shows the average duration of outbreaks as influenced by the number of agents, the presence of comorbidities, and different mitigation strategies (see Figure 3). Notably, the combination of mask-wearing and quarantine (both) consistently results in the shortest outbreak durations across both viruses, especially noticeable in larger populations. This implies that, especially in highly populated environments, integrating mitigation solutions can improve control measures in an effective manner.

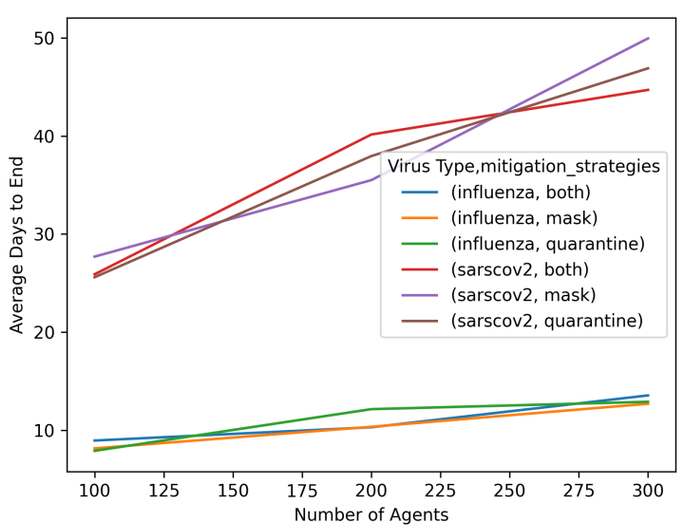

Figure 3: Average duration of outbreaks by number of agents and mitigation strategies.

6.4 Peak Infection Trends

The final graph depicts the average peak infection rates, highlighting differences between the number of agents and the applied mitigation strategy (see Figure 4). This visualisation shows that although the mitigation measures for both viruses flatten the peak infection curve, the effect is more noticeable in circumstances where both strategies are used. Notably, the combined strategy dramatically lowers peak infection rates for SARS-CoV-2, indicating its potential efficacy in real-life pandemic management.

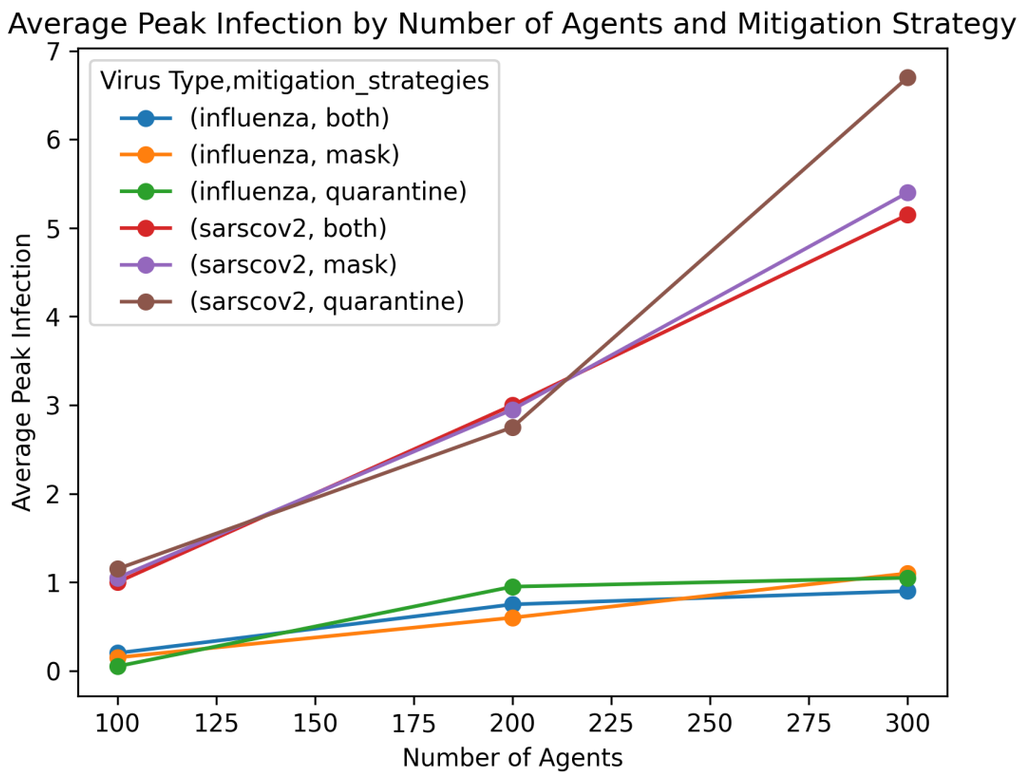

Figure 4: Average peak infection rates by mitigation strategy.

7 Discussion

The data visualisation highlights several important points:

7.1 Effectiveness of Combined Strategies

The best way to shorten the duration and peak of infectious outbreaks is to use both masks and quarantine. This combined approach proves significantly more effective than any single strategy alone, underscoring the synergistic effect of integrating multiple control measures.

7.2 Impact on Mortality

Mitigation strategies, particularly when combined, can effectively reduce mortality rates, especially in populations with higher comorbidities. This finding is crucial for public health planning, as it demonstrates the potential life-saving impact of comprehensive response strategies during pandemics.

7.3 Population Density

These techniques are especially important in more densely populated regions, as the impact of mitigation measures appears to scale with population density. High-density areas benefit substantially from strict mitigation measures, reflecting the enhanced risk of rapid virus transmission in such environments.

7.4 Anomalies and Considerations

- The similar total infection rates across different mitigation strategies suggest that while the spread can be slowed, the eventual reach of the virus may be similar unless strategies are continuously adapted and enforced. This points to the need for ongoing assessment and adjustment of public health responses to keep pace with evolving epidemiological insights.
- The comparison between masks and quarantine alone highlights issues related to compliance and enforcement that are not specifically addressed in the simulation but have practical consequences. Addressing these factors in future simulations could provide more realistic insights into the challenges of public health interventions.

These results emphasise how difficult it is to control infectious diseases and how adopting integrated strategies can have a big impact on how epidemic responses turn out. The models provide a starting point for additional study of certain policy initiatives and their real-world uses in public health planning.

8 Conclusion and Future Work

This project has illuminated significant differences in the transmission dynamics and impacts of Influenza A and SARS-CoV-2, particularly focusing on the influence of individual health parameters, such as comorbidities, in public health scenarios. The investigation has shown that mitigation strategies like mask wearing, quarantine, or their combination can significantly affect the progression and control of these infectious diseases within a simulated environment.

8.1 Significance of Findings

The consistent application of mitigation strategies across the population has shown little variation in the reduction of total infections, highlighting the need for widespread compliance to maximise public health outcomes. The data also emphasises the scalability of these interventions with population density, suggesting that more densely populated areas benefit significantly from rigorous public health measures.

8.2 Limitations

The study has limitations even if it offers useful data. The range of agent interactions was limited to public areas, which may have obscured transmission dynamics in other contexts, such as homes or offices. Furthermore, the simulation makes the assumption that all of the agent population behaves in the same way, which might not adequately represent the complexity and unique behavioural responses that exist in the actual world.

8.3 Future Research Directions

Subsequent studies could expand the environmental settings to include multiple interaction spheres, providing a more comprehensive model of community-wide transmission. The authenticity of the simulations could potentially be improved by incorporating behavioural models that represent different compliance with health measures. Furthermore, predicted accuracy and responsiveness might be further increased by utilising cutting-edge machine learning techniques to dynamically modify model parameters in response to real-time data.

8.4 Conclusion

Our simulations underscore the effectiveness of combined mitigation strategies (masks and quarantine) in markedly reducing the total number of infections, the severity of peak infection periods, and the duration of outbreaks, particularly in scenarios involving larger populations. This approach has proven especially effective against SARS-CoV-2, where the integration of both strategies substantially lowers peak infection rates, suggesting a potential roadmap for real-life pandemic management.

The findings suggest that while individual strategies like mask wearing or quarantine alone provide measurable benefits, their combined use synergistically enhances their effectiveness, reducing both the transmission rate and the potential mortality associated with the diseases. This is particularly critical in populations with higher comorbidity proportions, where the risk and severity of outcomes are greater. The study demonstrates that public health responses can be optimised by tailoring strategies to the specific characteristics of the population and the infectious agent involved.

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